

Sections: All CS and SE

Probability and Statistics (MT-2005)

Homework #01 (d)

Visualization of Bivariate Data

Question 1

Dataset: Programming Language Popularity Over Years

Using the given dataset, create multiple visualizations using Python (Matplotlib/Seaborn). Apply different visualization techniques to better understand the data trends.

Year	Python (%)	Java (%)	C++ (%)
2018	20	18	15
2019	25	17	14
2020	30	15	13
2021	40	12	12
2022	50	10	10

Table 1: Programming Language Popularity Over Years

Task 1: Multiple Charts

- Create **three separate line charts** for Python, Java, and C++ popularity over time.
- Use subplots to display them side by side.
- Interpret how each programming language's popularity has changed over the years.

Listing 1: Multiple Line Charts

```
import matplotlib.pyplot as plt
import numpy as np

years = [2018, 2019, 2020, 2021, 2022]
python_pop = [20, 25, 30, 40, 50]
java_pop = [18, 17, 15, 12, 10]
c_plus_plus_pop = [15, 14, 13, 12, 10]

fig, axes = plt.subplots(1, 3, figsize=(15,5))
axes[0].plot(years, python_pop, marker='o', color='b')
axes[0].set_title("Python Popularity")
```

```

axes[1].plot(years , java_pop , marker='o' , color='r')
axes[1].set_title("Java Popularity")
axes[2].plot(years , c_plus_plus_pop , marker='o' , color='g')
axes[2].set_title("C++ Popularity")
plt.show()

```

Task 2: Stacked Chart

- Create a **stacked area chart** to visualize the combined trends of Python, Java, and C++.
- Identify which programming language has shown the most growth and decline.
- Discuss key insights derived from the visualization.

Listing 2: Stacked Area Chart

```

fig , ax = plt.subplots(figsize=(10,6))
ax.stackplot(years , python_pop , java_pop , c_plus_plus_pop , labels=['Python'
ax.legend(loc='upper left')
plt.title("Stacked Area Chart of Programming Language Popularity")
plt.xlabel("Year")
plt.ylabel("Popularity (%)")
plt.show()

```